Assignment - 3

Section A - concept explanation type

1. Clocking methodology - It is a synchronization technique where operations occur based on clock signals, coordinating data movement between registers and functional units

2. Clock-Cycle - Smallest unit in processor operation, which determines speed of instruction execution.

3. Amdahl's law - states that speedup is limited by the non-parallelizable portion of a program.

4. Data path - consists of hardware components responsible for data processing including ALU, registers, multiplexers and buses.

5. Control unit - The control unit generates control signals that coordinate operations of ALU, registers & memory.

6. Single Cycle processor :- A single cycle processor executes the entire instruction in one clock cycle.

7. Multi-Cycle processor - divides instruction execution into multiple cycles with each stage executed in different clock cycles.

8. Instruction pipelining - pipelining in processors involves overlapping the execution of multiple instructions by dividing the instruction cycle into stages (eg IF, ID, EX, MEM, WB)

REDMI NOTE 6 PRO
MI DUAL CAMERA

9. Instruction level parallelism - ability to execute multiple instructions simultaneously in processor pipelines.

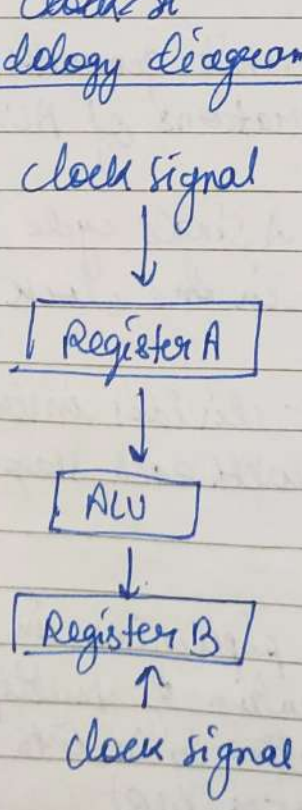
10. Data Hazards - a data hazard occurs when an instruction depends on the result of previous instruction that has not yet completed.

11. Control Hazards - due to branch instructions that affect the instruction flow in the pipeline.

12. Hazard Mitigation - pipeline Hazards can be mitigated using techniques such as forwarding, pipeline stalls branch prediction & instruction reordering.

Section-B

1. Clock Methodology Diagram.



clock signal synchronizes data transfers between registers and ALU.

2. Data path of a simple processor

memory → Registers → ALU → output

3. Instruction pipeline stages

Instruction 1: IF → ID → EX → MEM → WB
Instruction 2: IF → ID → EX → MEM → WB
Instruction 3: IF → ID → EX → MEM → WB
Instruction 4: IF → ID → EX → MEM → WB

IF → instruction fetch
ID - Instruction Decode.
EX - Execute.
MEM - Memory Access.
WB - write Back

4. Single cycle vs Multi cycle processor

Single cycle

|IF| ID |EX| MEM |WB| (All in one cycle)

Multicycle

cycle 1: IF
Cycle 2: ID
cycle 3: EX
Cycle 4: MEM
cycle 5: WB

REDMI NOTE 6 PRO
MI DUAL CAMERA

Section c - Case Based Analytical Answers

1. Data Hazards,
solution - forwarding, pipeline stalls.

2. speedup calculation using formula.

3. shorter clock cycles and better hardware utilization

4. Control hazard
- branch prediction techniques.

Section - D

1. instruction 1 : IF → ID → EX → MEM → WB
instruction 2 : IF → ID → EX → MEM → WB
instruction 3 : IF → ID → EX → MEM → WB

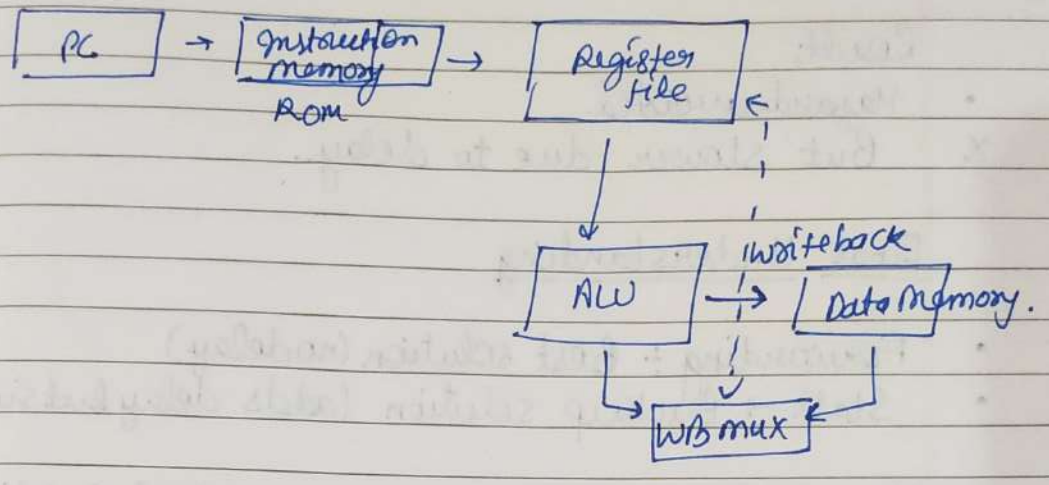
~~correct stage explanation and diagram clarity.~~

IF → ID → EX → MEM → WB

Execute (EX) - The ALU recieves two operands and a control signal specifying the operation (ADD, SUB, AND, SHL) The result and codes are latched into EX/MEM pipeline register at the clock edg.

REDMI NOTE 6 PRO
MI DUAL CAMERA

2. Simple CPU Datapath



3. instruction 1 : ADD R1, R2, R3
instruction 2: SUB R4, R1, R5

Data hazard occurs because instruction 2 depends on the result R1 produced by instruction 1 before it is completed

1. Using forwarding

Idea

Instead of waiting for ADD to store the result in the register, we directly pass (forward) the result from ALU to next instruction.

Pipeline View

Cycle :	1	2	3	4	5	6
ADD	IF	ID	EX	MEM	WB	
SUB		IF	ID	STALL	EX	MEM WB

SUB waits one cycle (stall) so that ADD completes and writes R1

Result

- Hazard avoided
- X But slower due to delay.

Final Understanding

- Forwarding ÷ Best solution (no delay)
- Stall → Backup solution (adds delay but simple)

If you want, I can give you a perfect 2-3 line exam answer for this so you can directly write it in your paper.

REDMI NOTE 6 PRO
MI DUAL CAMERA